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Research Statement

My research explores inequities in education and the factors that shape them. One strand of my work studies how teachers and learning environments shape gaps in achievement between boys and girls. A second strand studies teacher hiring and pay in Indian public education: specifically how quotas, credentials, and salary structures connect to teacher performance and student achievement. A third studies how students and school resources are sorted and allocated within local schooling markets. Across these projects, I work with large administrative panels, randomized experiments, and quasi-experimental designs, in settings from public schools in North Carolina and Texas to rural government schools and engineering colleges in India – measuring students' test scores, course grades, long-term outcomes like graduation and earnings, as well as psychosocial outcomes like anxiety, confidence, and beliefs, alongside teachers' credentials, performance, time use, and effort.

Published and Working Papers

Relative Skills in the Classroom: Teachers' Gender-Differentiated Impacts on Test Scores and Course Grades [Link to Paper]

Job Market Paper, Under Review

Girls outperform boys in teacher-assigned course grades by more than in standardized test scores. These gaps in grades expand as students progress through school, and predict gender gaps in long-term outcomes over and above what test score gaps predict. Teachers' impacts on test scores and course grades are both persistent and distinct from each other. In this paper, I ask whether these impacts differ between boys and girls – and if so, what explains the differences. Using administrative data from North Carolina, I find that fifth-grade teachers who improve test scores and those who improve course grades are largely different teachers – and they benefit different students. Teachers with high test score value-added disproportionately benefit girls' middle-school outcomes, and teachers with high grade value-added disproportionately benefit boys' outcomes. Teachers' grade value-added also raises boys' likelihood of remaining in school and graduating high school, with substantially smaller effects for girls. A simple model explains the asymmetry: if the skills measured by test scores and grades are imperfect substitutes, then teachers who are especially good at developing one of those skills have their largest impacts on the students who are relatively deficient in that skill. In turn, if boys and girls enter the classroom with different relative skill mixes, then teachers' gender-differentiated impacts arise without any role for teacher-student gender match or biased grading. Moreover, the same asymmetry should appear among students sorted by baseline skill mix regardless of gender – a prediction I confirm in the data. These findings link multidimensional teacher effectiveness with multidimensional gender gaps in student achievement, suggesting that teacher effectiveness is not a single attribute, and that its multidimensionality matters for which students benefit.

Affirmative Action, Faculty Productivity and Caste Interactions: Evidence from Engineering Colleges in India with Robert Fairlie, Saurabh Khanna, and Prashant Loyalka [Link to Paper]

Forthcoming at Journal of Political Economy Microeconomics

The central critique of affirmative action in hiring is an equity-efficiency tradeoff: workers hired through quotas have weaker credentials, and weaker credentials are presumed to mean lower productivity. However, evidence on the latter half of this assumption is mixed. We test it in the world's largest affirmative action program. Indian engineering colleges are required to reserve roughly half of faculty positions for candidates from disadvantaged castes and social classes. We collected data on faculty and students across a nationally representative sample of 50 colleges, including colleges that randomly assign students to classrooms and set course marks through externally graded exams. The first half of the assumption holds in our data: faculty hired through reservations have weaker credentials. The second half does not: we detect no gap in productivity across a wide range of student and faculty outcomes, including performance on standardized assessments, in-course assessments in immediate and follow-on courses, research output, and administrative service. On immediate course grades, students taught by reservation category faculty in fact perform slightly better, by about 1.3 to 1.5 percentile points – with comparable improvements for both general and reservation category students. Our findings also speak to the debate on affirmative action at an earlier stage: fewer than 2 percent of India's reservation-category population hold the required credentials for these positions, and most who do obtained them through reserved seats in student admissions. The faculty quota thus draws on a pool of potentially equally productive candidates who, absent affirmative action in student admissions, would largely not have held the qualifications to apply. The findings are evidence that faculty diversity can be achieved without compromising educational quality – in a highly technical field, across many universities, in the

world's most populous country.

Female Faculty Reduce Anxiety About STEM and Raise Academic Achievement Among Female Undergraduates with Robert Fairlie, Mridul Joshi, Saurabh Khanna, and Prashant Loyalka

Revise and Resubmit, Nature Communications

Draft available upon request

Female students in STEM perform better on academic assessments when taught by female faculty, and exposure to women in these fields also improves psychological outcomes for female students, by reducing anxiety and reshaping beliefs about who belongs in STEM. However, the performance effects and the psychological effects are rarely identified in a common, externally valid setting, and how the two are related is not well understood. We measure both sets of outcomes in one setting – 11 Indian engineering colleges wherein students are randomly assigned to instructors and graded through externally administered exams, with academic records paired to validated measures of STEM-related anxiety, confidence, and gender beliefs. Female students taught by female faculty earn higher course marks, by about 2.6 percentile points. Gains in course marks are concentrated among female students with lower prior achievement, confidence, or commitment to their field at entry. Furthermore, a 10 percentage point increase in the share of female instructors over the first two years closes 18 percent of the gender gap in academic skills observed at the beginning of the program, as measured through standardized tests. Increased exposure to female instructors also lowers female students' anxiety about mathematics and science, and male students' beliefs shift away from stereotypes about women's ability in STEM. These patterns are consistent with exposure to female faculty operating as a credible signal about who belongs in STEM – reducing psychological threat, with effects on skills that persist beyond the classroom.

Nonlinear Segregation Dynamics in Schooling Markets: The Case of Caste in India with Moumita Das, Shreya Dutt, and Kartik Srivastava [Link to paper]

Revisions in progress

Villages in rural India typically contain a handful of schools within short distances of each other, so families can move children between schools without relocating. We study whether these local schooling markets exhibit tipping points in caste composition: thresholds past which small changes in a school's composition trigger self-reinforcing sorting. Using near-universal administrative panel data on Indian schools over 12 years, we adapt the fixed-point threshold-estimation approach of Card, Mas, and Rothstein (2008) from residential segregation to multi-school markets. We find evidence that schooling markets do exhibit tipping behavior – and because sorting requires no relocation, segregation across schools within a village rises sharply at these thresholds while the village's own composition remains unchanged. Where the thresholds are identified depends on which caste boundary is in play: the boundary between upper-caste families and all disadvantaged groups tips at a different share than the boundary between intermediate groups and the most marginalized. Schools respond to tipping dynamics as well: inputs and public grants change discretely at the same thresholds, and government and private schools adjust differently.

Class Size Effects in Developing Countries: Longitudinal Evidence from India with Arushi Kaushik and Karthik Muralidharan

Draft coming soon

Pupil-teacher ratio norms are a central instrument of education policy in India: the Right to Education Act lowered the mandated ratio from 40:1 to 30:1, adding roughly 400,000 teachers at about $3.6 billion a year in salaries. Evidence is mixed on the effectiveness of pupil-teacher ratio reductions, especially in developing countries. We estimate the effect of the school-level pupil-teacher ratio on achievement in an annual panel of 500 rural government primary schools in the Indian state of Andhra Pradesh which were surveyed and tested over a five-year period. A 10 percent reduction in the ratio raises value-added scores by about 0.02 standard deviations – an effect that is consistent across a Maimonides IV and school fixed effects designs. The same designs diverge when estimated on achievement levels rather than gains, and the divergence is consistent with how much each design relies on the persistent component of variation in the pupil-teacher ratio within schools. This is a comparison that the class size literature has not made, and it is informative about what each design recovers. Priced as lifetime earnings, the gains exceed the cost of the additional teacher, with a benefit-cost ratio of about 1.8. In the same schools and on the same tests, preliminary results from our ongoing work on hiring a contract teacher and linking teacher pay to student performance delivered larger gains at lower cost.

Ongoing and Future Work

With Karthik Muralidharan and co-authors, I am working on two papers that draw on randomized experiments from the same Andhra Pradesh panel, each studying an alternative to the staffing policy that the class size paper benchmarks. The first asks whether teachers without professional training or civil-service protections can improve learning. These "contract" teachers are hired locally, serve on fixed-term renewable contracts, and are paid a fifth of a regular teacher's salary. In a government-implemented expansion, a randomly selected subset of APREStschools received an additional contract teacher. Preliminary results show that students in treated schools scored about 0.16 and 0.15 standard deviations higher in math and language than students in comparison schools in the first two years of the program. Furthermore, contract teachers were absent far less often than their tenured counterparts, putting their effectiveness at least on par with that of regular teachers – who are paid five times as much. The second studies teacher performance pay over five years – the longest horizon over which performance pay has been experimentally evaluated in any labor context. The first two years of results were published as Muralidharan and Sundararaman (2011, Journal of Political Economy); we extend the analysis through a full primary school cycle. Our preliminary results suggest that the gains from linking teacher salaries to a performance-based component hold over a full primary school cycle. Students who completed their entire primary schooling under individual teacher incentives scored about 0.54 and 0.35 standard deviations higher in math and language. The gains extend to science and social studies, subjects that were not linked to any incentives, and hold on questions testing conceptual understanding as well as rote learning, which is consistent with genuine additions to human capital rather than teaching to the test.

In ongoing work with Derek Rury, Sofia Shchukina, and Evan Bennett, I study whether success in high school sports spills over to students beyond the team, linking Texas administrative records that follow public high school students into college and the labor market with historical records of high school football performance. We use a school fixed effects design, and a regression discontinuity design comparing schools that narrowly won and narrowly lost playoff games. Preliminary estimates show no effect on test scores, but fewer disciplinary actions, higher college enrollment, and higher earnings among male students – a pattern consistent with success in a male-typed, high-status activity shifting boys' aspirations rather than their instruction.

In longer-term work with Karthik Muralidharan and Abhijeet Singh, I plan to use student panel data from Madhya Pradesh, Rajasthan, and Uttar Pradesh to estimate education production functions and characterize the misallocation of resources (especially teachers) in India's public education system. We plan to simulate expenditure-neutral reforms, such as consolidating small schools within local schooling markets and reallocating teachers across grades and schools, that can improve both equity and efficiency.